

and a third reviewer will resolve discrepancies. Two reviewers will critically appraise each included article using Joanna Briggs Institute critical appraisal tools, and a third reviewer will resolve discrepancies.

Data Synthesis: Descriptive analysis will be used to summarize the articles, the mobility and sleep outcome measures used, main results, and key findings across articles.

Conclusions: This systematic review will summarize the current state of the literature on how sleep impacts mobility in adults. It is anticipated that the findings will enhance our understanding of how sleep impacts mobility and the role of sleep as a target intervention to improve mobility.

Disclosures: C.S. is owner and CEO of Sleep Health Education, LLC. The other authors have nothing to disclose.

Keywords: Mobility, Rehabilitation, Scoping review, Sleep quality

Systematic & Meta-analytic Review Poster 2771656

Bi/multilingualism and Culturally Responsive Use of AAC: The Impact of Using All Linguistic Resources

Gianie Perez (Nemours Children's Hospital)

Objective(s): To understand the impact of bi/multiple language utilization and its related outcomes for effective augmentative and alternative communication (AAC) practice.

Data Sources: PubMed, CINAHL, Global Health, Cochrane Indexing terminology; pediatric bilingualism, pediatric intervention, multilingualism and inclusion, multilingual AAC use, multilingual speech-generating device, bilingual communicators (with, and) AAC, translanguaging, language activation, neurolinguistics, code-switching, caregiver training, outcomes in use of AAC.

Study Selection: Information was collected from articles retrieved from the abovementioned databases. Single subject case reviews were eliminated. The focus was on bilingual and multilingual speakers with limited inclusion of users of sign language (albeit indeed an alternative language); however, users of speech-generating devices were included. Of relevance to this research were the effect outcomes of therapy and consideration to support ongoing use of alternative communication approaches or devices.

Data Extraction: The review focused on research published in the last 20 years, with the exception of studies on mental health and bilingualism (eg, Cummins, 1989). Article selection and data extraction were relevant to the pediatric population.

Data Synthesis: The findings suggested that mastering a language and experimentation with words, their order, and sound system takes time. When children are exposed to multiple languages, there will be influences that carry across language systems, and overgeneralization is likely to occur. Similarly, when a bilingual/multilingual speaker has experienced language loss, language stimulation results in activation of all languages, facilitating communication. In summary, better performance is achieved with the promotion of language use in all forms, especially for multilingual speakers who use AAC as their primary function and when they are supported by their community. When AAC users, their caregivers, and their supportive staff become more comfortable using a specific device, it is more likely to be used in a variety of contexts. Thus, familiarity and acceptance increase and feelings of inadequacy and stigma decrease, resulting in the normalization of AAC utilization.

Conclusions: Qualitative research offered evidence that supported the notion that when multilingual AAC communicators are supported in the use of all known languages and their caregivers and support staff are trained and skilled in the use of the preferred devices, overall utilization increases. This results in furthering communicative independence and decreases caregiver reliance.

Disclosures: This author does not have anything to disclose as a conflict of interest.

Keywords: Alternative communication, Bilingualism, Multilingualism, Pediatrics, Rehabilitation

Systematic & Meta-analytic Review Poster 2751198

Examining Outcomes for Operative Versus Nonoperative Treatment of Pediatric Type II Supracondylar Humerus Fractures: A Systematic Review

Anthony Baumann (University Hospital),
Hudson Rogers, Kameron Shams, Omkar Anaspure,
Shiv Patel, R. Justin Mistovich

Objective(s): To examine various outcomes and complications in the literature associated with nonoperative versus operative care for type II supracondylar humerus fractures (SCHFs) in pediatric patients to guide surgeon decision making and optimize patient care.

Data Sources: PubMed, Web of Science, MEDLINE, and CINAHL were searched from database inception until February 7, 2024.

Study Selection: Articles were screened by 2 authors. Inclusion criteria were articles that examined pediatric patients, examined type II SCHFs, had English full-text, and compared operative versus nonoperative care. Seven articles met inclusion criteria for this study of 417 articles initially retrieved.

Data Extraction: Data extraction was performed by 3 authors. Data recorded included study type, study group by treatment type, patient population, diagnosis, number of patients, male/female status, mean patient age, treatment descriptions, follow-up time, and total and specific complications.

Data Synthesis: Numerous studies revealed similar values between treatment groups for carrying angle, Baumann's angle, elbow range of motion, and patient-reported function/satisfaction. Regardless of treatment, patients had low rates of avascular necrosis and loss of reduction, but nonoperative treatment was associated with a higher rate of residual deformity in the sagittal plane of the distal humerus (14.1% vs 0.0%; $P < .0001$). Failure of nonoperative treatment requiring conversion to operative treatment occurred in 8.9%-20.6% of patients. Factors associated with operative care were rotational deformity, varus/valgus malalignment, and shaft-condylar angle $< 30^\circ$.

Conclusions: Higher rates of sagittal deformity are associated with nonoperative management of type II SCHFs. However, the functional implications of this remain unclear as both operatively and nonoperatively treated patients have similar elbow range of motion, function, and satisfaction. Further studies should seek to identify specific anatomical parameters that identify those at risk of poor functional outcomes or are at higher risk of failing nonoperative management.

Disclosures: none.

Keywords: Orthopedics, Pediatrics, Rehabilitation, Supracondylar distal humeral fracture

Systematic & Meta-analytic Review Poster 2707549

How Do Pediatric Occupational Therapists Describe Their Process of Conducting Vision Evaluations?

Julia Mun (NYC Department of Education),
Victoria Syracuse

Objective(s): To describe the process of vision evaluations completed by pediatric occupational therapists, including those who work in schools and early intervention.

Data Sources: This descriptive qualitative research study used a series of 13 structured, open-ended interview questions via Zoom with eligible occupational therapist participants. Interviews conducted by investigators were recorded and transcribed. Participants checked transcriptions after interview completion, and investigators then coded the transcribed data manually using Excel.

Study Selection: Participants were included in this study if they met the following criteria: (1) occupational therapists currently practicing with the